

From Prime Splitting in the Eisenstein Integers to the Monster: A Computationally Verified Chain

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Abstract

We assemble, and verify computationally, a chain of classical results linking the splitting of rational primes in the Eisenstein integers $\mathbb{Z}[\omega]$ ($\omega = e^{2\pi i/3}$) to monstrous moonshine. The chain runs through: the theta series of the hexagonal (A_2) lattice and the quadratic character χ_{-3} ; the realization of that theta series as an automorphic induction from $\mathbb{Q}(\sqrt{-3})$ (a known instance of Langlands functoriality); the Borwein cubic theta functions and the genus-zero structure of $\Gamma_0(3)$; the Coxeter–Todd lattice K_{12} together with Eisenstein-ideal congruences; the order-three McKay–Thompson series of the Monster; and Ogg’s characterization of the supersingular primes as the prime divisors of the order of the Monster $\mathbb{M}$. *Every identity below is verified either as an exact q -series identity to a stated order or as a finite congruence/eigenvalue check over a stated range. We claim no new theorems.* The contribution is the explicit assembly of a single verified chain from a concrete elementary object to the Monster, and an honest delineation of what is theorem, what is verification, and where natural-looking inferences fail—in particular: the weight-6 level-3 newform is *not* of CM type; the prime 13 attached to K_{12} is basis-dependent; and the rank-12 Eisenstein lattices are unimodular and hence carry level-one (not level-three) theta series.

1 Introduction and scope

Fix $\omega = e^{2\pi i/3}$ and let $\mathbb{Z}[\omega]$ be the ring of Eisenstein integers, the ring of integers of $K = \mathbb{Q}(\sqrt{-3})$. A rational prime p splits, is inert, or ramifies in $\mathbb{Z}[\omega]$ according to the quadratic character χ_{-3} modulo 3:

$$\chi_{-3}(p) = \begin{cases} +1, & p \equiv 1 \pmod{3} \quad (\text{split}), \\ -1, & p \equiv 2 \pmod{3} \quad (\text{inert}), \\ 0, & p = 3 \quad (\text{ramified}). \end{cases}$$

This elementary datum is the seed. We trace it, by a sequence of *known* results, to the representation theory of the Monster simple group $\mathbb{M}$. Throughout, $q = e^{2\pi i\tau}$ with τ in the upper half-plane, η is the Dedekind eta function, and we write $\eta_3 = \eta(3\tau)$.

What this paper is, and is not. We make *no* claim of new mathematics. Each numbered statement is either a theorem in the literature (cited) or an elementary identity; in every case we report the explicit computational verification we performed (range of primes, order in q , etc.). The reason to record the chain is methodological: it shows that a single concrete picture—the distribution of primes by their fate in $\mathbb{Z}[\omega]$ —is bound, by verifiable identities, to objects of very

different appearance (lattices in 12 and 24 dimensions, Hauptmoduls, the Monster), and it isolates precisely *which* arithmetic survives each step and which does not. All computations were performed in exact arithmetic (Python/SymPy); see Section 9.

2 The character χ_{-3} and the A_2 theta series

Let $Q(a, b) = a^2 - ab + b^2$ be the norm form of $\mathbb{Z}[\omega]$ (equivalently $\frac{1}{2}$ the Gram form of the A_2 root lattice), and let $r(n) = \#\{(a, b) \in \mathbb{Z}^2 : Q(a, b) = n\}$. The theta series is $\Theta_{A_2}(\tau) = \sum_{n \geq 0} r(n)q^n$.

Proposition 1 (classical). *For $n \geq 1$, $r(n) = 6 \sum_{d|n} \chi_{-3}(d)$.*

This is standard (the class number of K is one, with six units). We verified $r(n) = 6 \sum_{d|n} \chi_{-3}(d)$ for all $1 \leq n \leq 2000$ by direct lattice enumeration, with no exceptions. The four qualitative features of the prime-splitting picture follow from the single formula: $r(p) = 12$ for split p , $r(p) = 0$ for inert p (the form never represents a prime $\equiv 2 \pmod{3}$; indeed Q takes only the residues $\{0, 1\}$ modulo 3), $r(p^2) = 6$ for inert p , and $r(3) = 6$.

Proposition 2 (classical). *Let $a(n) = r(n)/6 = \sum_{d|n} \chi_{-3}(d)$, the number of integral ideals of $\mathbb{Z}[\omega]$ of norm n . Then a is multiplicative and $\sum_{n \geq 1} a(n)n^{-s} = \zeta(s)L(s, \chi_{-3}) = \zeta_K(s)$.*

We verified multiplicativity of a on coprime pairs and the local Euler factors $a(p^k) = \sum_{i=0}^k \chi_{-3}(p)^i$ (giving $k+1$ at split p , 1 or 0 at inert p according to the parity of k , and 1 at $p = 3$) for all relevant $p^k < 2000$.

3 The rank-one Langlands instance: automorphic induction

Proposition 2 is the L -function form of automorphic induction from GL_1/K to $\mathrm{GL}_2/\mathbb{Q}$ for the trivial Hecke character; Θ_{A_2} is a weight-one form on $\Gamma_0(3)$ with nebentypus χ_{-3} , and this case is a theorem (Hecke's theory of theta series with characters; Langlands functoriality for $\mathrm{GL}_1 \rightarrow \mathrm{GL}_2$ over an imaginary quadratic field) [9, 11]. The sharp, eigenvalue-level statement is:

Observation 3 (verified). For every prime p , the normalized Hecke eigenvalue of Θ_{A_2} equals $a_p = 1 + \chi_{-3}(p)$: namely 2 at split p , 0 at inert p , and 1 at $p = 3$. This is the trace of Frobenius on the induced two-dimensional representation.

We verified $a_p = 1 + \chi_{-3}(p)$ for *all* primes $p < 4000$, with no exceptions, and confirmed the matching of local L -factors as in Proposition 2. We emphasize the direction of the statement: the automorphic form *encodes* the prime splitting of K ; it is not evidence that primes possess intrinsic higher-dimensional structure.

4 The Borwein cubic thetas and the $\Gamma_0(3)$ tower

Define the Borwein cubic theta functions through their standard η -quotient cubes [2]:

$$a(q) = \Theta_{A_2}, \quad b^3 = \frac{\eta(\tau)^9}{\eta(3\tau)^3}, \quad c^3 = 27 \frac{\eta(3\tau)^9}{\eta(\tau)^3}.$$

Theorem 4 (Borwein–Borwein [2]). $a^3 = b^3 + c^3$.

We verified $a^3 = b^3 + c^3$ as a q -series identity to order q^{40} . The Hauptmodul of $\Gamma_0(3)$ is $s = c^3/(27b^3) = (\eta(3\tau)/\eta(\tau))^{12}$, and the elliptic modular function j is a rational function of s of degree $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(3)] = 4$; we recovered

$$j = \frac{3^{18}s^4 + 4 \cdot 3^{14}s^3 + 2 \cdot 5 \cdot 3^9s^2 + 756s + 1}{s},$$

verified to order q^{25} . (Equivalently j is degree 4 over the function field of $\Gamma_0(3)$.)

5 The Coxeter–Todd lattice and Eisenstein-ideal primes

The Coxeter–Todd lattice K_{12} [5] is a 12-dimensional even $\mathbb{Z}[\omega]$ -lattice of determinant 3^6 ; its theta series lies in $M_6(\Gamma_0(3))$, a 3-dimensional space spanned by $\{b^6, b^3c^3, c^6\}$. Writing $f = (\eta(\tau)\eta(3\tau))^6$ for the unique normalized cusp form in $S_6(\Gamma_0(3))$, one has the elementary identities $f = b^3c^3/27$ and, since $a^3 = b^3 + c^3$, the pair b^3, c^3 are the roots of $T^2 - a^3T + 27f$.

Observation 5 (verified). $\Theta_{K_{12}} = b^6 + c^6 + 18f$, an integer combination with no occurrence of 13.

We verified this against the known representation numbers 1, 0, 756, 4032, 20412, 60480, 139860 of K_{12} . The prime 13 associated with K_{12} in the literature is an *Eisenstein-ideal* congruence prime of the Hecke algebra, not a feature of the lattice in the cubic-theta basis: in the basis of Eisenstein series $\{E_6(\tau), E_6(3\tau)\}$ and the Hecke-normalized cusp form one finds the cusp coefficient $-252/13$, and correspondingly

$$a_f(n) \equiv \sigma_5(n) - 27\sigma_5(n/3) \pmod{13} \quad (n \geq 1), \quad (1)$$

which we verified for all $1 \leq n \leq 1200$. Equation (1) is an instance of an Eisenstein-ideal congruence in the sense of Mazur [13]; its weight-12 level-one analogue is Ramanujan’s $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$ [16], which we likewise verified to $n = 1200$.

A governing quantity. Across weights k , the Eisenstein-ideal primes for $\Gamma_0(3)$ that we encountered are exactly the primes dividing the numerator of $(3^k - 1)B_k/(2k)$:

$$k = 6 : \text{numerator} = 13; \quad k = 12 : \text{numerator} = 73 \cdot 691 = 50443.$$

The factor $691 = \text{num } B_{12}$ is the level-one (Δ) prime, while 73 arises from $3^6 + 1 \mid 3^{12} - 1$. We verified by direct construction (Section 5.1) that 73 is a genuine Eisenstein-ideal prime of the unique level-3 weight-12 newform, and that 50443 is exactly the cusp denominator appearing when the orthogonal-sum lattice $A_2^{\oplus 12}$ is expanded in $M_{12}(\Gamma_0(3))$. This identity of quantities is consistent with, and we believe is subsumed by, the theory of Eisenstein ideals for $\Gamma_0(p)$ in higher weight [13]; we make no novelty claim.

5.1 The level-3 weight-12 newform and the status of 73

The space $S_{12}(\Gamma_0(3))$ is 3-dimensional; its oldspace is $\Delta(\tau), \Delta(3\tau)$ and its newspace is one-dimensional, spanned by a rational newform g (LMFDB label 3.12.a.a [12]). Constructing g as the T_2 -eigenvector with eigenvalue distinct from $\tau(2) = -24$, we found $a_g(2) = 78$ and verified

$$a_g(p) \equiv 1 + p^{11} \pmod{73} \quad \text{for all primes } p < 60, p \neq 3,$$

with $\gcd_p(a_g(p) - 1 - p^{11}) = 1971 = 3^3 \cdot 73$. Thus the Eisenstein-ideal primes of g are $\{3, 73\}$, and 73 is genuine, not a normalization artifact. The companion prime 691 belongs instead to the oldform Δ via Ramanujan’s congruence; the two primes together are the factors of the weight-12 Eisenstein numerator 50443.

6 The order-three classes of the Monster

By monstrous moonshine [4, 1], each conjugacy class g of $\mathbb{M}$ has a McKay–Thompson series T_g that is a Hauptmodul for a genus-zero group. The three classes of order three are realized over the $\Gamma_0(3)$ structure of Section 4; using sourced η -quotient forms [14, 3] we verified:

$$\begin{aligned} T_{3A} &= \frac{(h + 27)^2}{h} = \frac{a^6}{f} = q^{-1} + 42 + 783q + 8672q^2 + \cdots & [\Gamma_0(3)^+], \\ T_{3B} &= \left(\frac{\eta(\tau)}{\eta(3\tau)} \right)^{12} + 12 = q^{-1} + 54q - 76q^2 - \cdots & [\Gamma_0(3)], \\ T_{3C} &= \left(\frac{\eta(3\tau)}{\eta(6\tau)} \right)^8 + 256 \left(\frac{\eta(6\tau)}{\eta(3\tau)} \right)^{16} = q^{-1} + 248q^2 + \cdots & [3 \mid 3], \end{aligned}$$

where $h = (\eta(\tau)/\eta(3\tau))^{12}$. We verified each q -expansion against its published coefficients, the cubic-theta forms $T_{3A} = a^6/f = 27a^6/(b^3c^3)$ and $T_{3B} = 27b^3/c^3 + 12$ (to order q^{31}), and the moonshine consistency $c_g(1) = 1 + \chi_{196883}(g)$ in each case: $783 = 1 + 782$, $54 = 1 + 53$, and (for $3C$, whose q^1 coefficient is a gap) $0 = 1 + (-1)$. Two further verified relations:

- $T_{3C}(\tau)^3 = j(3\tau)$ (verified through q^{14} in $Q = q^3$): the $3C$ series is a cube root of j , the precise meaning of its “ $3 \mid 3$ ” structure.
- The level-three modular equation, uniformized by T_{3A} (verified to q^{11}):

$$j(\tau) + j(3\tau) = T_{3A}^3 - 126T_{3A}^2 + 2944T_{3A}, \quad j(\tau)j(3\tau) = T_{3A}^4 + 576T_{3A}^3 + 110592T_{3A}^2 + 7077888T_{3A}.$$

On the level-one side, the graded dimension of the Leech-lattice vertex operator algebra [8, 6] is $\Theta_\Lambda/\eta^{24} = j - 720$, and $196884 = 196560 + 324 = 1 + 196883$; we verified $\Theta_\Lambda/\eta^{24} = j - 720$ coefficientwise and the first head decompositions into Monster irreducibles.

7 Supersingular primes and Ogg’s theorem

Theorem 6 (Deuring [7]). *If an elliptic curve $E/\mathbb{Q}$ has complex multiplication by an order in an imaginary quadratic field K , then a prime p of good reduction is supersingular if and only if p is inert or ramified in K .*

For $E : y^2 = x^3 + 1$ ($j = 0$, CM by $\mathbb{Z}[\omega]$) this says: p is supersingular iff $p \not\equiv 1 \pmod{3}$. Computing the trace of Frobenius $a_p = p + 1 - \#E(\mathbb{F}_p)$ directly, we verified for all primes $5 \leq p < 400$ that $a_p \equiv 0 \pmod{p}$ (supersingularity) holds *exactly* when $p \equiv 2 \pmod{3}$. Thus the inert primes of Section 2 are the supersingular primes of the $j = 0$ curve.

Theorem 7 (Ogg [15]). *The primes p for which $X_0(p)^+ := X_0(p)/W_p$ has genus zero are exactly the prime divisors of $|\mathbb{M}|$.*

The order of the Monster factors as

$$|\mathbb{M}| = 2^{46} 3^{20} 5^9 7^6 11^{12} 13^3 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71,$$

so its prime support is $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 41, 47, 59, 71\}$. We computed the genus of $X_0(p)^+$ for all primes $p < 100$ from the genus formula for $X_0(p)$ together with the Riemann–Hurwitz count of fixed points of the Fricke involution W_p (via the class numbers $h(-4p)$ and, for $p \equiv 3 \pmod{4}$, $h(-p)$), and confirmed that genus zero occurs at *exactly* the prime divisors of $|\mathbb{M}|$ in that range. These genus-zero primes coincide with the supersingular primes of Theorem 6; cf. Ogg’s original observation [15].

8 What is and is not established

We can assert the following, each verified above:

1. The prime-splitting picture in $\mathbb{Z}[\omega]$ is the q -expansion of Θ_{A_2} , whose Hecke eigenvalues are $1 + \chi_{-3}(p)$; Θ_{A_2} is an automorphic induction from $\mathbb{Q}(\sqrt{-3})$ (Sections 2–3).
2. The same χ_{-3} organizes the $\Gamma_0(3)$ tower: the cubic thetas with $a^3 = b^3 + c^3$, the Coxeter–Todd lattice ($\Theta_{K_{12}} = b^6 + c^6 + 18f$), the order-three McKay–Thompson series ($T_{3A} = a^6/f$, $T_{3C}^3 = j(3\tau)$), and the level-three modular equation (Sections 4–6).
3. The inert primes are the supersingular primes of the $j = 0$ CM curve, and the supersingular (equivalently Ogg/genus-zero) primes are exactly the prime divisors of $|\mathbb{M}|$ (Section 7).

We explicitly do *not* assert, and the computations actively caution against, the following:

1. *That primes possess intrinsic higher-dimensional structure that “descends.”* The direction of every result above is that automorphic and lattice objects *encode* prime arithmetic, not the reverse. The free commutative monoid $\bigoplus_p \mathbb{Z}_{\geq 0}$ on the primes is a genuine infinite-dimensional discrete object, and χ_{-3} colours its coordinate axes (the ideal-counting function factors over axes); but the A_2 *norm form*—the metric and packing structure—does not transfer to this flat exponent lattice.
2. *That the weight-six newform $f = (\eta\eta_3)^6$ has CM by $\mathbb{Q}(\sqrt{-3})$.* A direct check shows $a_f(2) = -6 \neq 0$, so f does not vanish at inert primes; its link to χ_{-3} is the congruence (1), not CM.
3. *That the prime 13 is intrinsic to K_{12} .* In the cubic-theta basis $\Theta_{K_{12}} = b^6 + c^6 + 18f$ contains no 13; the prime is an Eisenstein-ideal feature of the Hecke basis.
4. *That the rank-12 Eisenstein ($\mathbb{Z}[\omega]$ -) lattices extend the level-three (χ_{-3}) thread to dimension 24.* By the Hentschel–Krieg–Nebe classification [10] these five lattices ($3E_8, 4E_6, 6D_4, 12A_2$, Leech) are even *unimodular*, hence have level-one weight-12 theta series governed by 691, not 13; χ_{-3} leaves the theta series upon gluing to unimodularity. The genuine dimension-24 χ_{-3} -thread runs instead through the non-unimodular 3-modular lattice $A_2^{\oplus 12}$ (Section 5).

None of the identities in this paper is claimed as new. Their interest is the verified coherence of the chain and the sharp separation of what each step does and does not carry.

9 Computational methods

All verifications used exact arithmetic in `Python` with `SymPy`. The A_2 data were obtained by direct lattice enumeration over $\{(a, b) : 0 < Q(a, b) \leq N\}$ with the exact bound $|a| \leq \lceil (2/\sqrt{3})\sqrt{N} \rceil$ (so that no representation near the diagonal directions is truncated). Modular and theta q -expansions were computed as exact integer or rational power series: η -products by iterated multiplication by $(1 - q^{mn})$, quotients by power-series inversion, and Hecke operators by $(T_p f)(n) = a_f(pn) + p^{k-1}a_f(n/p)$. The level-3 weight-12 newform was extracted as a T_2 -eigenvector on the basis $\{\Delta(\tau), (\eta\eta_3)^{12}, \Delta(3\tau)\}$ of $S_{12}(\Gamma_0(3))$. Class numbers $h(-d)$ were computed by enumerating reduced primitive positive-definite binary quadratic forms, and the genus of $X_0(p)^+$ by the standard genus formula for $X_0(p)$ together with Riemann–Hurwitz for the degree-two Fricke quotient. Point counts on $y^2 = x^3 + 1$ over $\mathbb{F}_p$ used the Legendre symbol directly. Coefficient data were cross-checked against the LMFDB [12] and the OEIS [14] where applicable (sequences A030197 for T_{3A} and A007245 for T_{3C}).

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